Department of Information Technology
Academic Year 2023– 2024(Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT
Course Code & Title: CS3362 & Foundations of Data science
Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic: Unit 4 / Aggregation and Grouping**
- **Course Outcome:** The student will be able to know the concept of Aggregation and Grouping in data science.
- **Topic Learning Outcome:** TLO9.
- **Activity Chosen: Cross Word Puzzle**
- **Justification:**
  - The concept of Aggregation and Grouping is one of the important topics for making data analysis. To understand the level of students this activity is planned.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
- In the previous day class was taken about Aggregation and Grouping.
- Ask the students to prepare well for the topic.
- Printout sheet was given to students to fill the puzzle.
- The Question and Answer for the topic as shown in Figure 1.
- The sample presentation of the students Ms.S.Sobana, P.Ananthi, Kavi Praba G and Devadharshini K as shown in Figure 2 and 3.
- **CO – PO / PSO mapping:**

CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO9**	**PO10**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO4	3	3	2	2	2	1	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO9**	**PO10**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	3	2	2	2	1	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of Aggregation and Grouping for data science problems	Identify the need of Aggregation and Grouping for developing applications	Able to design solutions for implementing Aggregation and Grouping to find relation between data	Able to analyze the relationship between data and provide valid conclusions on data using Aggregation and Grouping	Able to perform analysis and group of data	Based on the analysis people can communicate to a group of peoples	Ability to recognize the need for data Aggregation and Grouping in real world applications	Exhibit knowledge in Aggregation and Grouping techniques to provide solution for complex computational problems	Exhibit knowledge in Aggregation and Grouping techniques in the field of IoT, Cloud Computing and Data Analytics to provide IT solutions.	Exhibit knowledge in Aggregation and Grouping techniques in software projects to deliver a quality product

- **Images / Screenshot of the practice:**

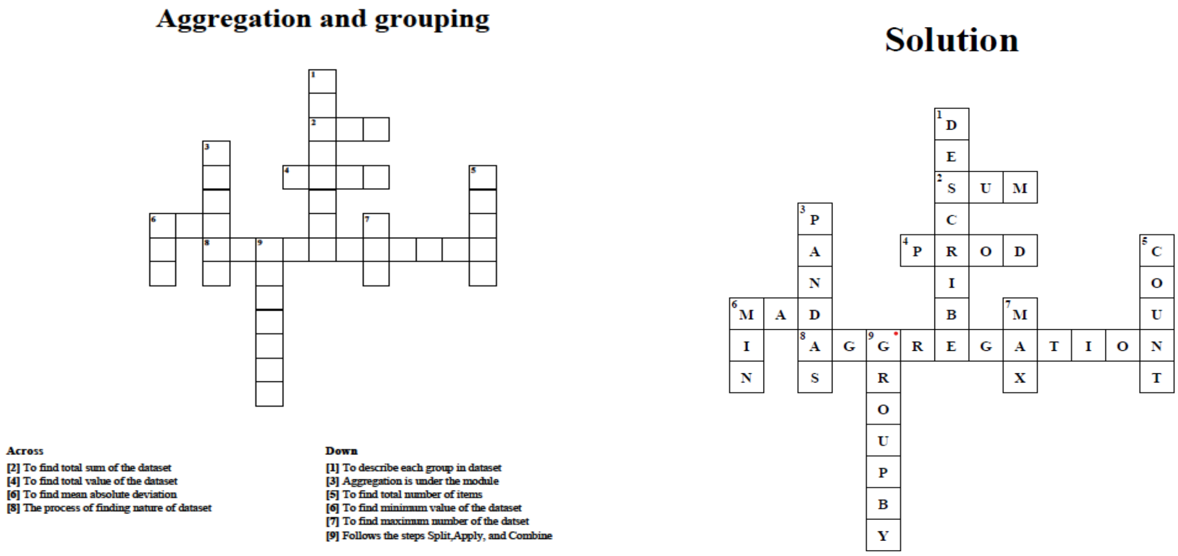

Figure1: Questions and answer of Cross word puzzle

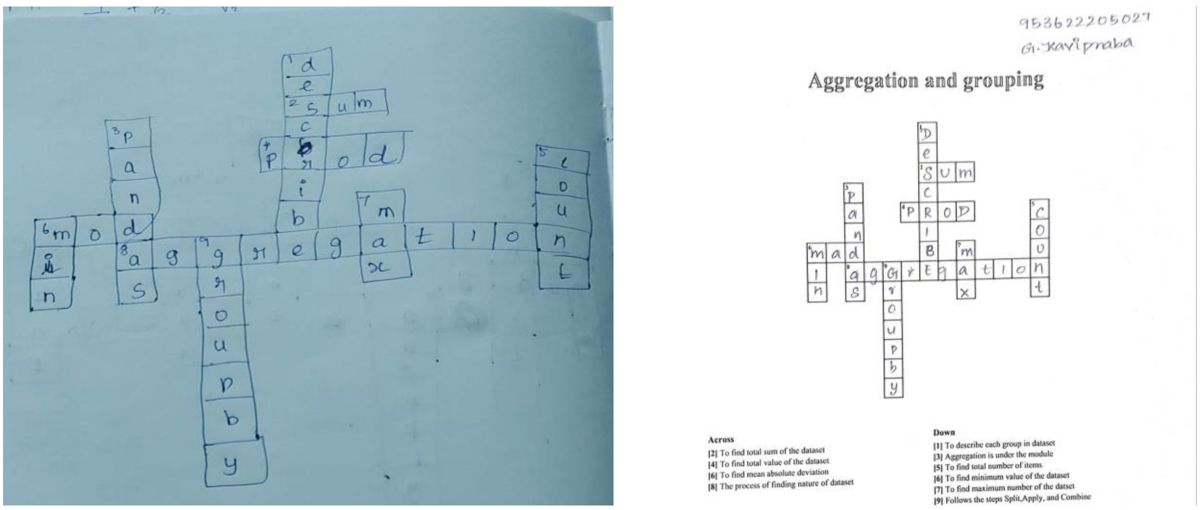

Figure 2: Answers given by the students Sobana S and Kavi Praba G

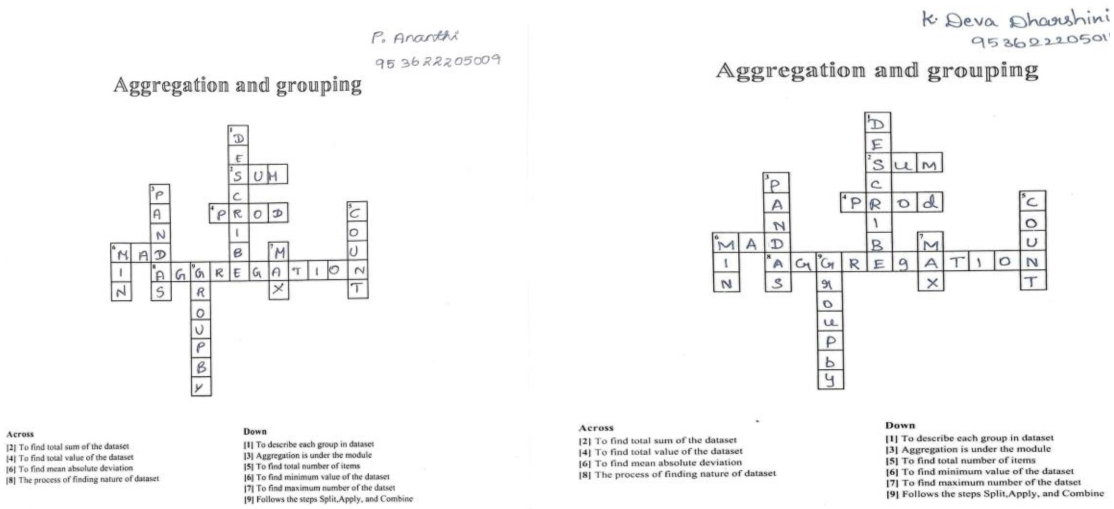

Figure 3: Answers given by students Ananthi. P and Deva Dharshini K

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - The students are actively participated in the activity.
    - The students have got the chance to examine their comprehension of the topic of Aggregation and Grouping.
    - With the help of this activity, students can gain knowledge about grouping and aggregation as well as how to analyze data and make conclusions.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - The students understand well about the median, mean and mode and based on that how to make analysis on data.
    - The students and teacher can interact well during this activity.
    - Students' fear decreased when they talked to their peers about it.
    - This activity helps the students to think logically.
    - Each student will be fully aware of how to apply grouping and aggregation in data analysis.
  - ***Challenges faced in implementation:***
    - Some students do not know the answer for the given questions. The answers are discussed in the next class.
    - The activity was planned for 10 Minutes, but it takes 15 minutes because the students doesn't know the answer.

References:

- https://www.ritrjpm.ac.in/images/computer-science/
- https://www.ritrjpm.ac.in/images/computer-science/2023-2024/IP/CS3391_GSP_23_24_CWP.pdf

Signature of Faculty Member **HOD**